

- Q.32 Discuss the Raoult's Law, Dalton's Law.
- Q.33 Write any 5 differences between distillation and extraction process.
- Q.34 Write effect of impurities on crystal formation.
- Q.35 Explain Mier's saturation theory.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the construction and working of agitated tank crystallizer.
- Q.37 Write short note on any two:-
- Adsorption isotherms
 - Steam Distillation
 - Relative Volatility
- Q.38 Note : Attempt both Questions
- List the names of Plates used for distillation column and explain any one with neat sketch.
 - Draw neat diagram and explain packed column for Distillation.

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Roll No.

5th Sem / Chem, P & P Subject:- Mass Transfer - II

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Distillation is generally used for cement industries.
- True
 - False
- Q.2 In case relative volatility is equal to 1 then distillation is used
- Steam
 - Azeotropic
 - Fractional
 - None of the above
- Q.3 RO is an application of
- Adsorption
 - Membrane separation
 - absorption
 - None of these
- Q.4 Which of the following process can accelerate Distillation?
- Temperature
 - Pressure
 - Both
 - None
- Q.5 The term Entrainer is used in which distillation
- azeotropic
 - flash
 - steam
 - fractional

- Q.6 Solvent extraction is basically known as
- Gas-liquid extraction
 - Liquid-liquid extraction
 - Liquid-solid extraction
 - None of above
- Q.7 Crystallization is based on the difference of
- Melting point
 - boiling point
 - pressure
 - solubility
- Q.8 Separating of a component from a liquid by solvent is named as
- Extraction
 - leaching
 - crystallization
 - distillation
- Q.9 Distillation is a _____ operation
- Solid-liquid
 - Gas-liquid
 - Liquid-Liquid
 - Gas-Gas
- Q.10 Liquid-Liquid mixture is separated with solvent extraction by adding _____ solvent.
- Soluble
 - Insoluble
 - Partially soluble
 - none

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Explain the term R.O.
- Q.12 Name the equipment used for leaching operation.
- Q.13 Solubility curve is a relation between _____ and _____

- Q.14 What is Filtration.
- Q.15 The other name for Q line is _____.
- Q.16 Write any one example of crystallization process.
- Q.17 Write full form of BET.
- Q.18 Write formula for reflux ratio?
- Q.19 What is loading?
- Q.20 Name any one mass transfer process?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write concept of crystallization in brief.
- Q.22 Express in Mathematically Expression relative Volatility and write its significance.
- Q.23 Write down a Note on Leaching Process.
- Q.24 Explain mixer settler with construction and working.
- Q.25 Draw a well labeled diagram of Swenson walker crystallizer and Write its's any 2 advantages and applications.
- Q.26 Explain any 2 types of membrane.
- Q.27 Explain various types of adsorption.
- Q.28 Write about the baffle tube crystallization.
- Q.29 Write the methods for achieving super saturation.
- Q.30 Write about concept of Dialysis.
- Q.31 Write industrial application of adsorption.